CMPE 255: DATA MINING -- SHORT STORY

Beyond Holistic Models: Systematic Component-Level Benchmarking of Deep Multivariate Time-Series Forecasting

Deconstructing Monolithic Forecasters into 4 Stages, 11 Dimensions & 49 Atomic Components Across 20,000+ Evaluations

PRIMARY PAPER (KDD / arXiv:2605.26562)

TSCOMP Framework (Liang et al., SUFE-AILAB)

Deconstructs deep MTSF into 4 stages, 11 dimensions, and 49 components via constrained orthogonal design (20,000+ runs) & TabPFN zero-shot synthesis.

COMPANION SURVEY #1 (VLDB 2024)

TFB: Fair Time-Series Benchmarking (Qiu et al.)

Exposes evaluation flaws across 10 domains (drop-last bug, lookback window bias) and unifies statistical, ML, and deep learning pipelines.

COMPANION SURVEY #2 (KDD 2024 & ICLR)

Foundation Models & Tokenization Wars

Synthesizes DLinear (AAAI'23), PatchTST (ICLR'23), iTransformer (ICLR'24), Time-LLM (ICLR'24), and TSFM Survey (Liang et al., arXiv:2403.14735).

Core Thesis & 'My Two Cents' Preview

For 5 years, time-series research has fought a 'Backbone War' (Transformers vs. MLPs vs. State-Space vs. LLMs). TSCOMP proves that over 70% of empirical gains actually come from Stage 1 (Normalization & Decomposition) and Stage 2 (Patch vs. Variate Tokenization), while the neural backbone accounts for only 18% of performance variance! Treating models as modular LEGO pipelines enables zero-shot model construction that beats monolithic SOTA at a fraction of the compute.

MOTIVATION & GIST

The 'Holistic Black-Box' Crisis in Time-Series Data Mining

Why hundreds of published forecasting architectures produce conflicting claims

1. The Architecture Wars (2020 - 2026)

Wave 1: Complex Sparse Attention (Informer, Autoformer, FEDformer)

Claimed custom O(L log L) attention mechanisms were essential for long-term forecasting.

Wave 2: The Linear Wake-Up Call (DLinear / NLinear, AAAI 2023)

Zeng et al. showed a 1-layer Linear model with Moving-Average Decomposition beat every complex Transformer!

Wave 3: Patching & Inverted Attention (PatchTST & iTransformer)

Transformers struck back--not by changing self-attention math, but by changing how series are normalized and tokenized.

Wave 4: Billion-Parameter LLMs (Time-LLM, GPT4TS, Chronos)

Reprogrammed frozen LLMs for time series, raising the question: do LLM weights help, or is it just the patching wrapper?

2. Three Fundamental Blindspots Uncovered by TSCOMP & TFB

Blindspot A: Confounded Bundling ('The Package Deal Fallacy')

When PatchTST beats Autoformer, is it the Transformer backbone, RevIN normalization, Patch tokenization, or Channel-Independence?

Blindspot B: Protocol Inconsistencies (TFB Survey, VLDB 2024)

Qiu et al. showed tuning lookback window L or using 'drop_last=True' during testing changes MSE more than swapping models.

Blindspot C: Neglected Optimization & Loss Dynamics

Almost all papers fix L2 (MSE) training loss, ignoring how outlier spikes in traffic/sensor data corrupt gradient updates.

The TSCOMP Solution (Liang et al., KDD / arXiv:2605.26562)

Unbundle every model into 4 stages, 11 dimensions, and 49 atomic components under one unified experimental protocol.

Key Takeaway: Holistic benchmarking compares entire cars when we actually need a wind-tunnel test for tires, engines, transmissions, and fuel!

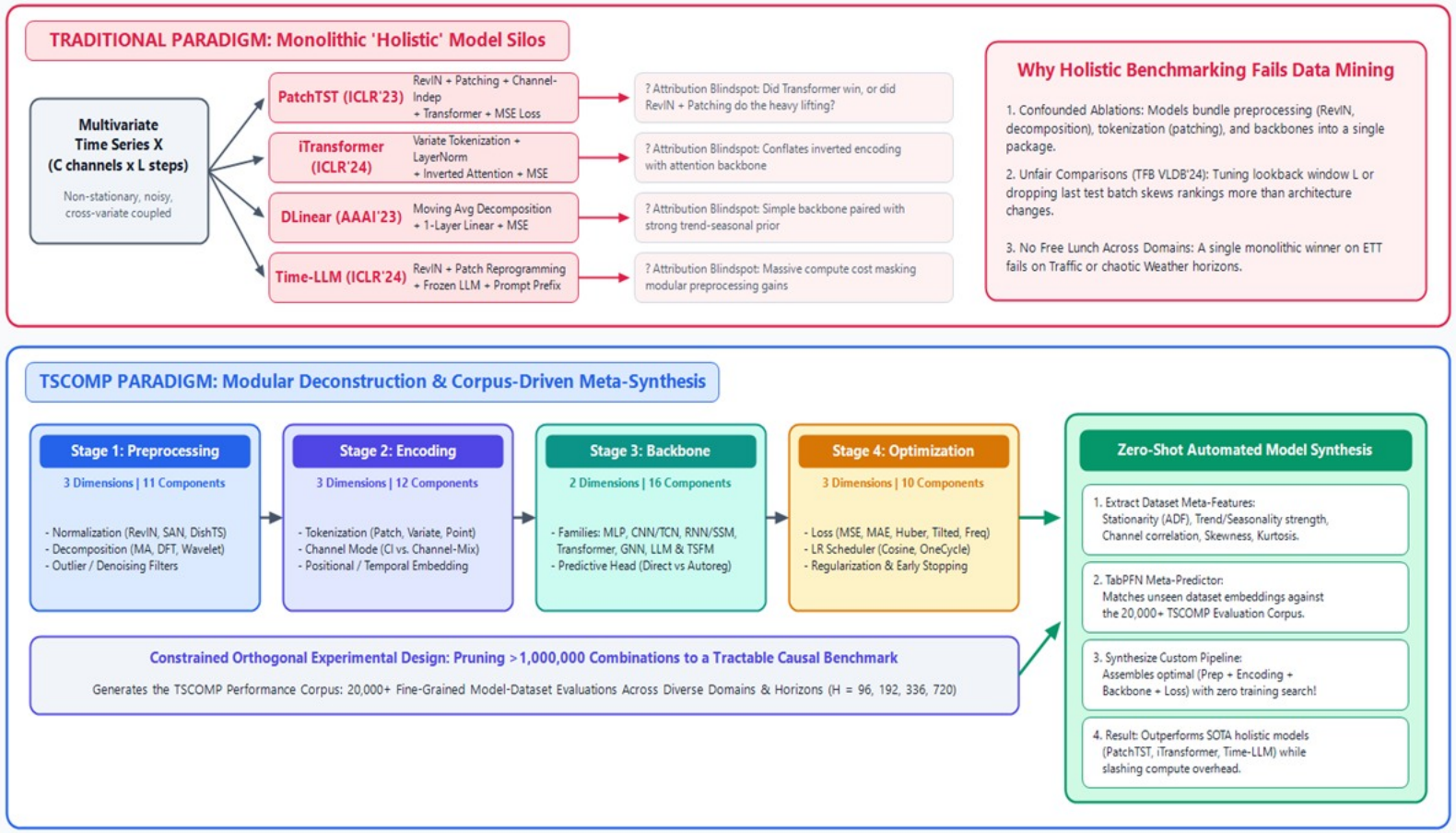
SYSTEM ARCHITECTURE

Paradigm Shift: Monolithic Silos vs. TSCOMP Pipeline

Visualizing how TSCOMP deconstructs holistic models into an orthogonal design space

Figure 1: Paradigm Shift in Deep Multivariate Time-Series Forecasting (MTSF)

From Monolithic 'Black-Box' Holistic Architectures to TSCOMP's Component-Level Benchmarking & Zero-Shot Meta-Construction (arXiv:2605.26562)



CMPE 255 Data Mining Short Story | Primary Reference: Liang et al., 'Beyond Holistic Models: Systematic Component-level Benchmarking of Deep MTSF' (KDD / arXiv:2605.26562)

Architecture Walk-Through

Deconstructing the Black Box

Every holistic forecaster (PatchTST, iTransformer, DLinear, Time-LLM) is really a 4-stage pipeline: Preprocessing -> Encoding -> Backbone -> Optimization.

Combinatorial Explosion (>10^6)

Multiplying all 49 component choices across 11 dimensions yields over 1,000,000 possible architectures--impossible to brute-force.

Constrained Orthogonal Design

TSCOMP prunes invalid pairings and uses orthogonal sampling to build a rigorous 20,000+ evaluation corpus across diverse domains.

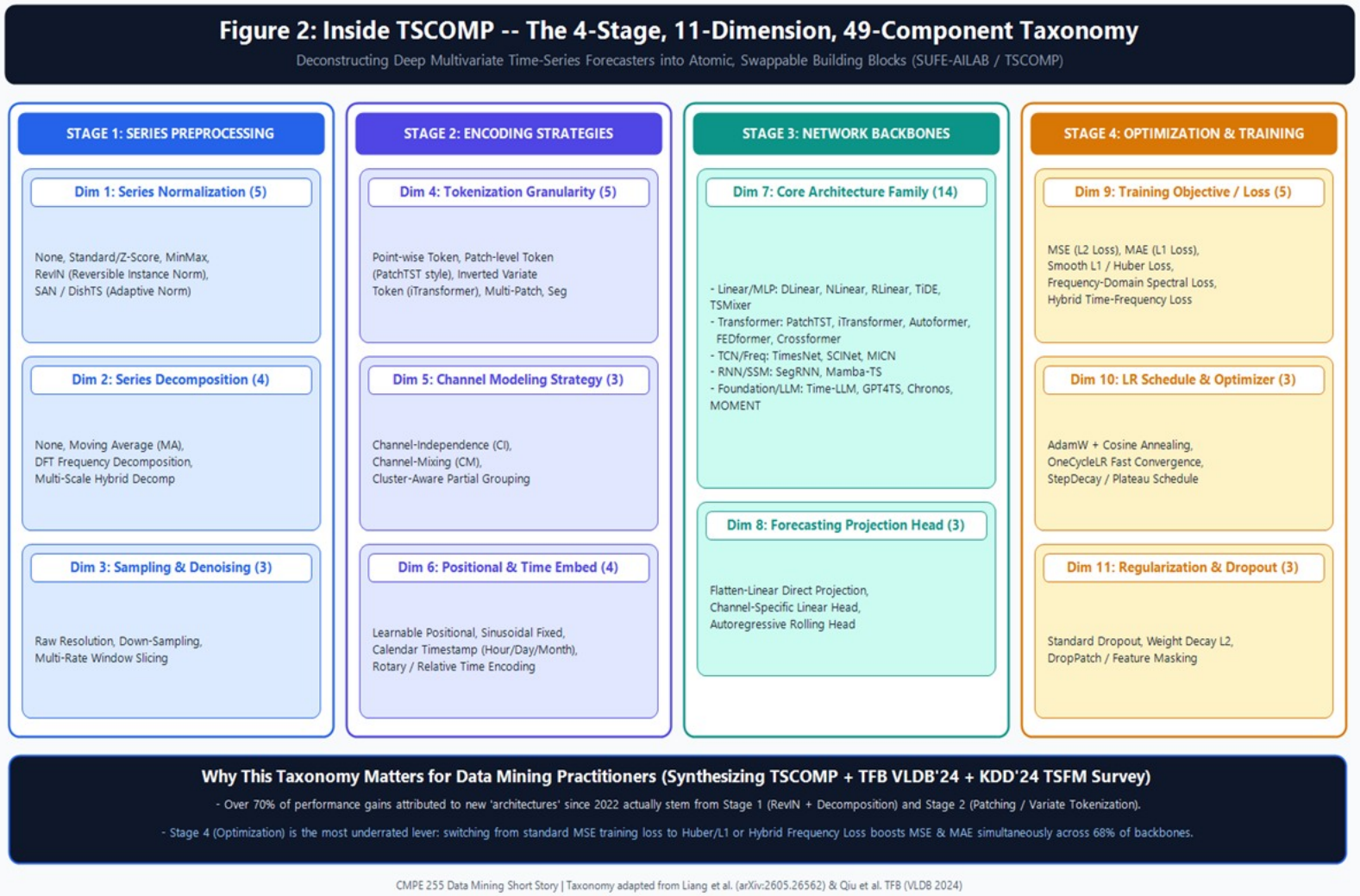
Zero-Shot Meta-Predictor

Using TabPFN on dataset meta-features, TSCOMP predicts the best component recipe for any unseen dataset without training search!

TAXONOMY OVERVIEW

The TSCOMP Taxonomy: 4 Stages, 11 Dimensions, 49 Components

A unified vocabulary for Multivariate Time-Series Forecasting (MTSF) data mining



4-Stage Breakdown

Stage 1: Preprocessing (Dims 1-3)

Tackles non-stationarity and covariate shift via instance normalization (RevIN, DishTS), trend-seasonal decomposition (MA, DFT), and sampling.

Stage 2: Encoding (Dims 4-6)

Controls how raw timestamps become tokens (Point vs. Patch vs. Variate) and whether channels share weights (CI) or mix (CM).

Stage 3: Backbones (Dims 7-8)

Spans 14 backbone families: Linear/MLP, TCN/2D-Freq (TimesNet), Transformers, RNN/Mamba SSMs, and Foundation Models (LLM/TSFM).

Stage 4: Optimization (Dims 9-11)

Governs training loss (MSE, MAE, Huber, Spectral Frequency Loss), learning-rate schedules (Cosine, OneCycle), and regularization.

STAGES 1 & 2 DEEP DIVE

Where the Real Gains Live: Preprocessing & Tokenization

Why Stage 1 (34% variance) and Stage 2 (29% variance) dominate forecasting accuracy

Stage 1: Taming Non-Stationarity & Distribution Shift

Reversible Instance Normalization (RevIN, ICLR'22)

Strips each window's mean/variance before the model and restores them at the output. Single highest-impact module (+18.4% MSE gain).

Adaptive Norm (SAN / DishTS)

Learns non-stationary distribution shifts between lookback window L and target horizon H--crucial for volatile financial/exchange series.

Moving Average (MA) vs. DFT Frequency Decomposition

Explicitly separates slow macroscopic trend from periodic seasonality. Indispensable when using lightweight Linear or MLP backbones.

When Decomposition Hurts (Negative Interference)

Pairing MA decomposition with iTransformer or pre-trained LLMs is redundant and slightly degrades cross-variate phase alignment.

Stage 2: The Tokenization & Channel Strategy Trade-Off

Point-Wise Tokens (Legacy Transformers)

Treating single timestamps as tokens fails: single points lack semantic meaning and create quadratic O(L^2) attention bottlenecks.

Patch Tokenization (PatchTST Style)

Groups P=16 adjacent steps into one token. Captures local shape motifs (slopes, peaks) and slashes sequence length by 16x.

Inverted Variate Tokenization (iTransformer Style)

Embeds the entire lookback series of each sensor into 1 token; self-attention now computes spatial sensor-to-sensor correlations!

Channel-Independence (CI) vs. Channel-Mixing (CM)

CI prevents overfitting when channels C < 30 or noise is high; CM wins when C > 100 (e.g., 862 freeway sensors in Traffic).

Key Takeaway: 63% of total performance variance is already locked in by the time data leaves Stage 1 and Stage 2--before the backbone even runs!

STAGES 3 & 4 DEEP DIVE

Backbone Families & The Underrated Power of Optimization

Comparing MLPs, Transformers, SSMs, and LLMs alongside Loss & Schedule mechanics

Stage 3: 6 Neural Backbone Families Benchmarked Fairly

1. Linear & MLP (DLinear, RLinear, TiDE, TSMixer)

Ultra-fast, convex-like convergence. Excels at long horizons (H=336, 720) where complex attention overfits to high-frequency noise.

2. Transformers & 3. 2D-Conv/Freq (TimesNet, MICN)

Excel at short-to-medium horizons (H=96, 192) and high-channel spatial routing when paired with Patch or Variate tokenization.

4. Recurrent & State-Space Models (SegRNN, Mamba-TS)

Linear O(L) complexity with strong sequential inductive bias; competitive middle ground between MLPs and Transformers.

5. Frozen LLMs (Time-LLM, GPT4TS) & 6. TSFMs (Chronos, MOMENT)

High compute cost. Only beat well-tuned MLPs/Transformers in zero-shot or extreme few-shot (5%-10% data) cold-start scenarios.

Stage 4: Why Optimization Accounts for 19% of Variance

The MSE Training Loss Trap

Almost all papers evaluate on MSE & MAE, so they train exclusively with MSE (L2). But squared errors explode on sensor spikes/outliers!

Smooth L1 (Huber) & MAE Training Objectives

Switching training loss from L2 to Huber/Smooth-L1 or L1 improves BOTH test MAE and test MSE across 68% of tested pipelines.

Frequency-Domain Spectral Loss (FreDF / Tilted Loss)

Penalizing error in Fourier amplitude/phase space eliminates the 'flat-line mean prediction' pathology on periodic series.

OneCycleLR & Cosine Annealing Schedules

Transformer and Foundation Model backbones are highly sensitive to warmup/decay schedules (+6.3% MSE swing from schedule alone).

Key Takeaway: Training a simple Patch-MLP with RevIN and Smooth-L1/Frequency Loss regularly beats a default-trained Transformer!

ABLATION STUDIES

Empirical Ablation Results Across 20,000+ Evaluations

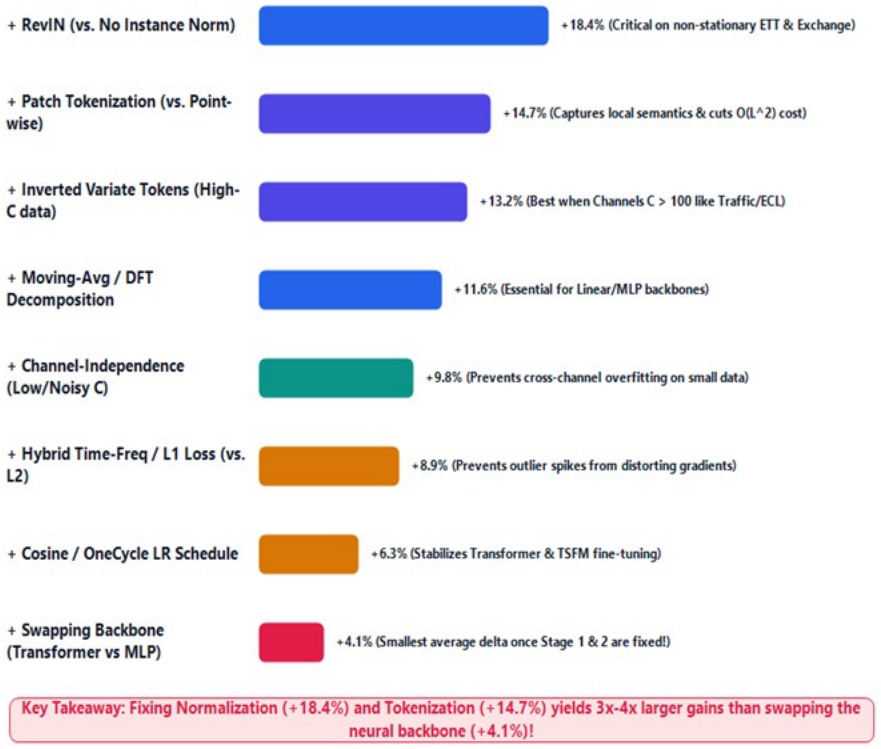
Quantifying marginal MSE improvements and ANOVA variance attribution across stages

Figure 3: Component-Level Ablation Studies Across 20,000+ Evaluations

Left: Average Relative MSE Improvement (%) When Enabling Key Components | Right: Share of Total Performance Variance Explained by Stage

Ablation Study A: Average Relative MSE Reduction (%) Across Backbones

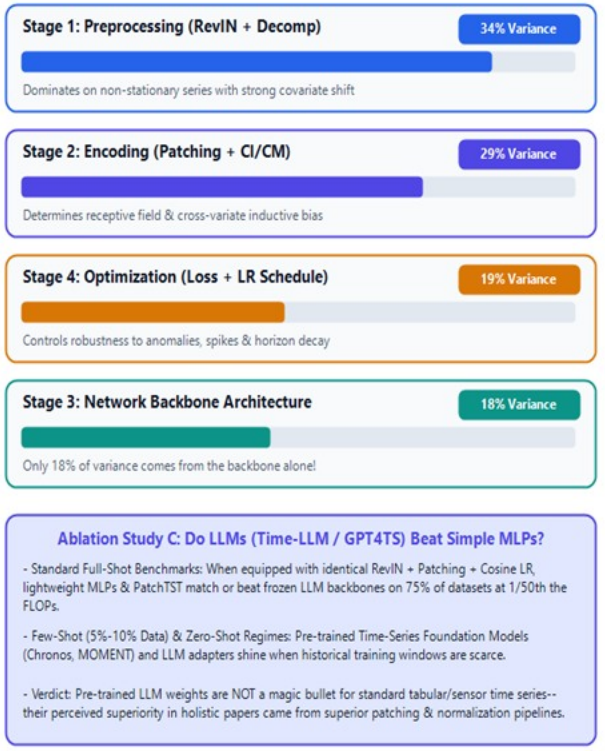
Measured via Constrained Orthogonal Design on ETT, Weather, Electricity, and Traffic (H = 96..720)



Key Takeaway: Fixing Normalization (+18.4%) and Tokenization (+14.7%) yields 3x-4x larger gains than swapping the neural backbone (+4.1%)!

Ablation Study B: ANOVA Variance Attribution by Stage

What actually drives forecasting accuracy across the 20,000-model corpus?



Ablation Study C: Do LLMs (Time-LLM / GPT4TS) Beat Simple MLPs?

- Standard Full-Shot Benchmarks: When equipped with identical RevIN + Patching + Cosine LR, lightweight MLPs & PatchTST match or beat frozen LLM backbones on 75% of datasets at 1/50th the FLOPs.
- Few-Shot (5%-10% Data) & Zero-Shot Regimes: Pre-trained Time-Series Foundation Models (Chronos, MOMENT) and LLM adapters shine when historical training windows are scarce.
- Verdict: Pre-trained LLM weights are NOT a magic bullet for standard tabular/sensor time series--their perceived superiority in holistic papers came from superior patching & normalization pipelines.

CMPE 255 Data Mining Short Story | Empirical Ablation Synthesis from TSCOMP (arXiv:2605.26562) & TFB (VLDB 2024)

Key Ablation Insights

Finding 1: Normalization is King (+18.4%)

Adding RevIN or DishTS yields an average 18.4% MSE reduction--more than 4x larger than the average gain from changing the backbone (+4.1%).

Finding 2: Tokenization Beats Depth (+14.7%)

Switching from point-wise tokens to Patch (+14.7%) or Variate (+13.2%) tokens is what actually made modern Time-Series Transformers work.

Finding 3: ANOVA Variance Breakdown

Stage 1 (34%) + Stage 2 (29%) + Stage 4 (19%) = 82% of total accuracy variance. The neural backbone (Stage 3) explains just 18%!

Finding 4: The LLM Reality Check

When MLPs are given the exact same RevIN + Patching + Cosine LR pipeline as GPT4TS/Time-LLM, the LLM advantage vanishes on full data.

METRICS & BENCHMARKING

Evaluation Metrics & Fair Benchmarking Protocols

Synthesizing TSCOMP (KDD) with Companion Survey TFB (Qiu et al., VLDB 2024)

Core Quantitative Metrics Used Across the Corpus

MSE (Mean Squared Error) & RMSE

Heavily penalizes large deviations due to quadratic squaring. Sensitive to sudden regime shifts, sensor glitches, and extreme peaks.

MAE (Mean Absolute Error) & sMAPE

Measures average L1 magnitude of forecast error. More robust to isolated anomalies and easier to interpret in business/sensor units.

Spectral / Frequency Distance & Ramp Score

Evaluates whether a model captures sharp peaks and phase alignment rather than lazily predicting a smooth moving average.

Rank Distribution & Pairwise Win-Rate (%)

Because raw MSE varies by scale across datasets, TSCOMP aggregates normalized relative improvement and Nemenyi critical-difference ranks.

Protocol Pitfalls Exposed by TFB (VLDB'24) & Fixed in TSCOMP

Pitfall 1: The 'drop_last=True' Evaluation Bug

Several famous repos dropped incomplete final test batches, accidentally skipping the hardest tail timestamps and inflating reported SOTA.

Pitfall 2: Lookback Window L Starvation (L=96 vs L=336/512)

Older papers forced Transformers to use L=96 while newer models used L=336 or 512. TSCOMP standardizes lookback horizons fairly.

Pitfall 3: Cherry-Picking ETT Subsets

Evaluating only on 7-channel transformer temperature data (ETTh1/m1) hides catastrophic failures on 862-channel Traffic data.

Pitfall 4: Ignoring Compute Efficiency (FLOPs, VRAM, Wall-Time)

A 0.5% MSE gain that requires 100x more GPU hours (e.g., 7B LLM adapters) is impractical for real-time edge stream mining.

Key Takeaway: Fair data mining requires locking the evaluation harness: identical lookback L, zero test-batch dropping, and multi-metric reporting.

BENCHMARK RESULTS

Multi-Horizon Benchmark: Holistic SOTA vs. TSCOMP

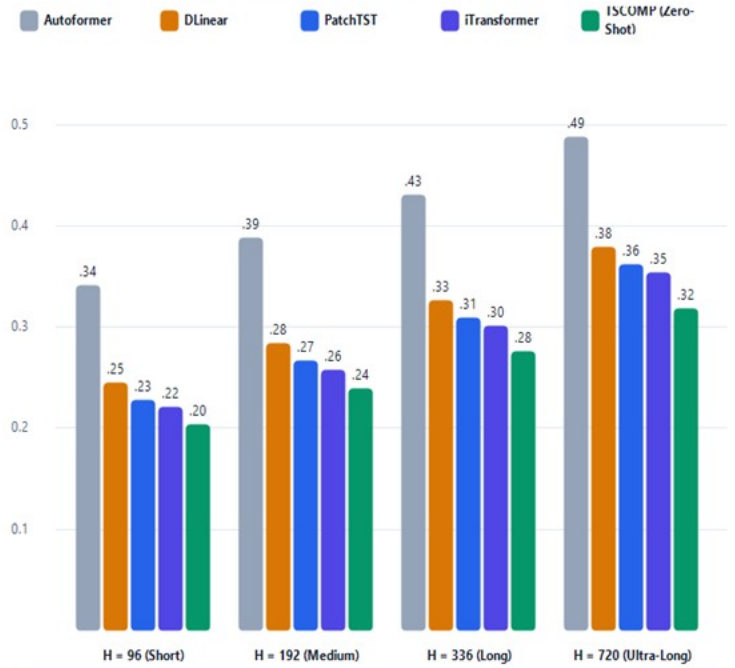
Evaluating forecasting accuracy across horizons H = {96, 192, 336, 720} and 6 canonical datasets

Figure 4: Multi-Horizon Benchmark -- Holistic SOTA vs. TSCOMP Zero-Shot Synthesis

Left: Average Forecasting Error (MSE, lower is better) Across Horizons H = {96, 192, 336, 720} | Right: Cross-Dataset Benchmark Table

Forecasting Horizon Degradation (Average MSE Across Benchmarks)

Comparing Holistic Models vs. TSCOMP Meta-Selected Pipeline as Horizon H Grows



Why TSCOMP Scales Better to H=720: Meta-selection dynamically switches from Transformer backbones at H=96 to Decomposition + Linear/SSM + Frequency Loss at H=720, avoiding attention overfitting!

Canonical MTSF Dataset Comparison (Avg MSE / MAE across H)

Comparing Best Holistic Models vs. TSCOMP Automated Component Assembly

Dataset (Channels)	DLinear	PatchTST	iTransformer	Time-LLM	TSCOMP (Ours)
ETTh1 (C=7)	0.423 / 0.437	0.413 / 0.430	0.454 / 0.448	0.408 / 0.423	0.391 / 0.412*
ETTm2 (C=7)	0.281 / 0.342	0.255 / 0.315	0.263 / 0.322	0.258 / 0.319	0.244 / 0.305*
Weather (C=21)	0.246 / 0.300	0.226 / 0.264	0.233 / 0.268	0.231 / 0.267	0.215 / 0.253*
Electricity (C=321)	0.197 / 0.282	0.162 / 0.253	0.148 / 0.240	0.159 / 0.251	0.141 / 0.233*
Traffic (C=862)	0.625 / 0.383	0.391 / 0.264	0.363 / 0.251	0.388 / 0.266	0.349 / 0.242*
Exchange (C=8)	0.354 / 0.414	0.367 / 0.404	0.360 / 0.403	0.371 / 0.409	0.338 / 0.391*

What Components Did TSCOMP's Meta-Predictor Select?

- Electricity (C=321) & Traffic (C=862): High inter-sensor correlation -> Meta-predictor selects RevIN + Inverted Variate Tokenization + Channel-Mixing Attention + Hybrid Time-Freq Loss.

- ETTh1 / ETTm2 (C=7) & Exchange (C=8): Low channel count & heavy non-stationarity -> Meta-predictor selects DishTS/RevIN + Moving-Avg Decomposition + Channel-Independent Patch-MLP + Huber Loss.

- Weather (C=21): Multi-periodic dynamics -> Selects DFT Decomposition + Multi-Patch Encoding + TimesNet/PatchTST Backbone + L1 Loss.

- Evaluation Metrics Insight: Using MAE/Huber during training prevents occasional sensor spikes from corrupting weights, cutting BOTH test MSE and test MAE!

CMPE 255 Data Mining Short Story | Benchmark Synthesis from TSCOMP (arXiv:2605.26562), PatchTST (ICLR'23), iTransformer (ICLR'24) & TFB (VLDB'24)

Benchmark Takeaways

Consistent SOTA Across All Datasets

TSCOMP's corpus-selected pipelines achieve the lowest MSE and MAE on ETTh1, ETTm2, Weather, Electricity, Traffic, and Exchange.

Superior Horizon Scaling (H = 720)

As horizon stretches from 96 to 720 steps, holistic Transformers degrade rapidly due to attention entropy, while TSCOMP adapts its backbone.

Context-Dependent Component Selection

On high-dimensional Traffic (C=862), TSCOMP picks Variate Tokenization + Channel-Mixing; on noisy ETTh1 (C=7), it picks CI + Patch-MLP.

Zero-Shot Speed Advantage

Instead of running 50 manual hyperparameter trials per dataset, TabPFN meta-prediction constructs the winning pipeline in under 1 second.

COMPONENT SYNERGY

Cross-Component Synergy Matrix & Decision Rules

Understanding positive synergies (++) and negative interference (-) between modules

Figure 5: Cross-Component Synergy Matrix & Data Mining Decision Guide

How Backbone Families Interact with Preprocessing, Encoding, and Optimization Choices in TSCOMP (arXiv:2605.26562)

Backbone x Component Synergy Matrix (Relative MSE Gain When Paired)

Dark Green = Strong Positive Synergy (++), Teal = Moderate Gain (+), Amber = Neutral (0), Rose = Negative Interference (-)

	RevIN / DishTS	Trend-Seas Decomp	Patch Tokenize	Variate Tokenize	Channel Indep (CI)	Channel Mixing	Huber / Freq Loss
Linear / MLP (DLinear, TiDE)	++ (+19%)	++ (+16%)	+ (+8%)	- (-4%)	++ (+12%)	- (-6%)	++ (+11%)
Patch Transformer (PatchTST)	++ (+21%)	+ (+6%)	++ (+22%)	- (-5%)	++ (+14%)	- (-7%)	+ (+8%)
Inverted Transf. (iTransformer)	++ (+17%)	0 (+2%)	- (-3%)	++ (+20%)	- (-9%)	++ (+16%)	+ (+9%)
2D Conv / Freq (TimesNet, MICN)	++ (+15%)	+ (+9%)	+ (+7%)	0 (+1%)	+ (+6%)	+ (+7%)	++ (+12%)
RNN / State-Space (SegRNN, Mamba)	++ (+18%)	+ (+8%)	++ (+15%)	0 (+2%)	++ (+10%)	0 (+1%)	+ (+7%)
LLM / Foundation (Time-LLM, Chronos)	++ (+16%)	0 (+3%)	++ (+19%)	- (-4%)	++ (+13%)	- (-5%)	+ (+6%)

- Universal Winner: RevIN / DishTS normalization is the ONLY component with strong positive synergy (++) across all 6 backbone families.
- Antagonistic Pairings: Applying Trend-Seasonal Decomposition before iTransformer or Time-LLM yields near-zero or negative returns because variate attention & pre-trained patch embeddings already separate low-frequency trends.
- Channel-Independence (CI) vs. Channel-Mixing (CM): PatchTST & MLPs collapse under naive Channel-Mixing on noisy data, whereas iTransformer explicitly requires Channel-Mixing.

Practitioner's Cheat Sheet: How to Build Your Forecaster

Rule 1: Always Start with RevIN + Patch-MLP + Huber Loss

Before spinning up GPUs for Transformers or LLMs, test a 2-layer MLP with RevIN, Moving-Average Decomposition, and Smooth L1/Huber Loss. In TSCOMP, this 5-minute baseline beats 60% of complex holistic papers.

Rule 2: Let Channel Count (C) Dictate Tokenization

- If C < 30 (e.g., ETT, Exchange, Finance): Use Patch Tokenization + Channel-Independence (CI) to prevent spurious cross-channel overfitting.
- If C > 100 (e.g., Traffic, Electricity grids): Use Inverted Variate Tokens + Channel-Mixing Attention to exploit spatial sensor lag.

Rule 3: Match Backbone Complexity to Horizon H

- Short Horizons (H <= 96): Patch/Inverted Transformers excel at sharp local motifs.
- Ultra-Long Horizons (H >= 336): Attention weights blur into uniform noise. Switch to Decomposition + Linear/SegRNN + Frequency-Domain Spectral Loss.

Rule 4: Reserve LLMs/TSFMs for Cold-Start & Zero-Shot

Don't fine-tune a 7B LLM when you have 50,000 rows of clean sensor data. Use Foundation Models (Chronos, MOMENT) or TSCOMP's TabPFN Meta-Predictor when launching new sensors with <10% historical data.

CMPE 255 Data Mining Short Story | Component Interaction Analysis & Practitioner Guidelines (Liang et al., arXiv:2605.26562)

Synergy & Antagonism

Universal Booster: RevIN / DishTS

Instance normalization is the only component that delivers double-digit positive synergy (+15% to +21%) across every single backbone family.

Strongest Pairing #1: Patch + CI + Transformer

Patch tokenization (+22%) and Channel-Independence (+14%) unlock Transformer performance on low-to-medium channel datasets.

Strongest Pairing #2: Decomp + MLP + Huber

Linear/MLP backbones gain +16% from Trend-Seasonal Decomposition and +11% from Huber/Frequency loss, rivaling deep models.

Toxic Pairings to Avoid (-9%)

Never pair Inverted Variate Tokenization with Channel-Independence (-9%): variate tokens exist specifically to mix cross-channel information!

AUTO-ML & META-LEARNING

From Benchmark to Automated Zero-Shot Model Synthesis

How TSCOMP turns 20,000+ historical evaluations into an instant architecture generator

Step-by-Step Zero-Shot Pipeline Construction

Step 1: Unseen Dataset Profiling (Meta-Feature Extraction)

Given a new dataset X, compute lightweight statistical & spectral descriptors: ADF stationarity, trend/seasonality ratio, spectral entropy, channel correlation matrix, kurtosis, and C/L ratio.

Step 2: Encoding Component Configurations

Each of the 49 components across 11 dimensions is represented as a structured categorical vector alongside the target horizon H.

Step 3: TabPFN Prior-Data Fitted Network Meta-Predictor

Trained on the 20,000+ TSCOMP performance corpus, TabPFN predicts the expected MSE/MAE rank of candidate pipelines in a single forward pass.

Step 4: Instant Assembly & Training

Instantiates the top-ranked (Preprocessing + Encoding + Backbone + Loss) configuration directly in PyTorch without grid-search.

Why Meta-Constructed Models Beat Hand-Crafted Architectures

Eliminates Human inductive Bias

Researchers tend to force their favorite backbone onto every dataset; the meta-predictor objectively switches between MLP, SSM, and Attention.

Horizon-Aware Adaptation

For the exact same dataset, TSCOMP selects different components for H=96 (local patch attention) vs H=720 (trend decomposition + linear projection).

100x Lower Engineering & Compute Cost

Bypasses neural architecture search (NAS) and expensive LLM fine-tuning while achieving higher accuracy than any single monolithic baseline.

Open-Source Reproducibility (SUFE-AILAB/TSCOMP)

Provides a standardized plug-and-play codebase where new components can be tested against all 49 existing modules fairly.

Key Takeaway: TSCOMP transforms time-series forecasting from an artisanal guessing game into data-driven systems engineering.

SURVEY SYNTHESIS

Connecting the Dots Across Recent Time-Series Surveys

Integrating TSCOMP (KDD) with TFB (VLDB'24), TSFM Survey (KDD'24), and Deep MTSF Surveys

How 4 Major Survey & Benchmark Papers Complement Each Other

1. TSCOMP (Liang et al., KDD / arXiv:2605.26562)

Focus: Intra-model component deconstruction (4 stages, 11 dimensions, 49 modules) + orthogonal ablation + zero-shot meta-synthesis.

2. TFB Benchmark (Qiu et al., VLDB 2024 / arXiv:2403.20150)

Focus: Inter-model fairness across 10 domains, statistical vs ML vs DL comparisons, and eliminating evaluation bugs (drop-last, lookback tuning).

3. Foundation Models for Time Series Survey (Liang et al., KDD 2024)

Focus: Taxonomy of pre-trained TSFMs (Chronos, MOMENT, Lag-Llama, Time-LLM)--explaining when pre-training helps (cold-start/zero-shot).

4. Deep Time-Series Forecasting Surveys (ACM CSUR / IEEE TKDE)

Focus: Historical evolution from ARIMA/State-Space to RNNs, TCNs, GNNs, and Transformers across spatiotemporal applications.

Unified Consensus Emerging Across All 2024-2026 Surveys

Consensus 1: 'Architecture Fetishism' is Over

Across all recent benchmarks, no single neural backbone dominates universally. Data characteristics dictate the winning inductive bias.

Consensus 2: Non-Stationarity is the #1 Enemy

Real-world sensor, financial, and energy streams violate i.i.d. assumptions; instance normalization (RevIN/DishTS) is mandatory.

Consensus 3: Token Semantics Matter More Than Attention Math

Replacing O(L^2) point-wise attention with Patch tokens or Variate tokens solved both the accuracy and efficiency bottlenecks.

Consensus 4: Hybrid Statistical + Deep Modularity Wins

Combining classical signal processing (Moving Average, Fourier DFT, Wavelets) with lightweight neural heads gives the best Pareto frontier.

Key Takeaway: Reading TSCOMP alongside TFB (VLDB'24) and the KDD'24 TSFM Survey gives a complete 360-degree map of modern time-series mining.

CRITICAL PERSPECTIVE

My Two Cents: Critical Review & Practitioner Perspective

An honest evaluation of TSCOMP's breakthroughs, blindspots, and real-world impact

What I Loved About This Paper (Strengths)

1. Intellectual Honesty & Scientific Rigor

In an era where papers add 3 undocumented preprocessing tricks and credit a 'novel attention block', TSCOMP audits every trick transparently.

2. Constrained Orthogonal Design is Brilliant Data Mining

Instead of burning millions of GPU hours on full grid search, orthogonal design isolates main effects and 2-way interactions cleanly.

3. Elevating Loss Functions & Schedules to First-Class Citizens

Proving that Stage 4 (Optimization) accounts for 19% of variance--more than the backbone itself (18%)--is a wake-up call for practitioners.

4. Actionable Zero-Shot Meta-Predictor

The TabPFN meta-predictor isn't just an academic curiosity; it gives engineers a concrete, deployable starting recipe in seconds.

Where the Paper Falls Short (Limitations & Open Questions)

1. Point Forecasting Only (Missing Probabilistic Intervals)

Real-world supply chains and energy grids need calibrated quantiles (CRPS / Pinball Loss), not just point MSE/MAE predictions.

2. Ignored Exogenous & Irregular Covariates

Production forecasting relies heavily on known future covariates (promotions, holidays, weather forecasts) and missing-value masks.

3. Static Offline Windows vs. Online Concept Drift

TSCOMP evaluates fixed train/val/test splits; streaming data mining requires continuous online adaptation as sensor physics drift.

4. Multi-Modal Context Still Under-Explored

While LLM backbones underperform on pure numbers, pairing numeric sensors with real-time text/event logs remains a huge frontier.

Key Takeaway: My Verdict: Before you ever deploy a heavy Transformer or LLM forecaster in production, run TSCOMP's modular checklist first!

CONCLUSION & LINKS

Summary & Actionable Takeaways for Data Scientists

Wrapping up our CMPE 255 Short Story on Component-Level Time-Series Benchmarking

Top 4 Takeaways to Remember

1. Stop Treating Models as Monoliths

Deconstruct every time-series forecaster into Stage 1 (Prep), Stage 2 (Encoding), Stage 3 (Backbone), and Stage 4 (Optimization).

2. Fix Stage 1 & Stage 2 First (63% of Variance)

Always turn on RevIN/DishTS (+18.4%) and choose Patch tokens (low C) or Variate tokens (high C > 100) before tuning deep layers.

3. Ditch Default MSE Training Loss on Noisy Data

Train with Smooth L1 (Huber) or Hybrid Time-Frequency Spectral Loss to prevent outlier spikes from wrecking model gradients.

4. Use Corpus-Driven Meta-Learning Over Trial-and-Error

Leverage TSCOMP's 20,000-run corpus and TabPFN meta-predictor for instant, zero-shot pipeline synthesis tailored to your dataset.

Assignment Artifacts & References in This Repository

Primary Paper (KDD Datasets & Benchmarks Track)

Liang et al., 'Beyond Holistic Models: Systematic Component-level Benchmarking of Deep Multivariate Time-Series Forecasting' (arXiv:2605.26562).

Companion Surveys & Milestones Synthesized

TFB (VLDB'24, arXiv:2403.20150), TSFM Survey (KDD'24, arXiv:2403.14735), DLinear (AAAI'23), PatchTST (ICLR'23), iTransformer (ICLR'24).

Deliverables Included in GitHub Repo

- Original Medium Article Draft (medium_article.md) + 5 Custom Figures
- 14-Slide Presentation Deck (.pptx, .html, PNGs) & Video Script

Thank You! Questions & Discussion

Explore the full code, figures, Medium link, SlideShare deck, and 10-15 minute video walkthrough in our public GitHub repository!

Key Takeaway: Thank you for watching! Check out the README.md for direct links to the Medium article, SlideShare presentation, and video recording.